

The Hedge Resilience Index (HRI) White Paper

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Preface

The AI era is driving a fundamental structural transformation in the business landscape. Legacy operations — built on decades of accumulated product portfolios, customer relationships, and operational models — face mounting pressure from decelerating growth and structural disruption. Meanwhile, AI-driven businesses are progressively opening new avenues for revenue generation. The central question is not whether AI-native ventures will replace traditional operations, but whether available capital reserves, strategic time horizon, and organizational capabilities are sufficient to complete the transition before legacy revenues enter sustained decline.

The Hedge Resilience Index (HRI) is designed to address a fundamental question: do the cash flows and time generated by legacy operations provide sufficient runway to secure a viable position in the emerging AI economy? This is not a financial ratio exercise — it is an assessment of an organization's capacity to sustain its transformation through the transition period.

HRI is designed to provide a consistent and comparable analytical framework. Drawing on seven core indicators calibrated to sector-specific characteristics, it offers a quantitative assessment of an enterprise's AI transformation resilience and progress. The resulting data indicates whether an organization is currently in the critical transition phase, or approaching — and potentially crossing — the inflection point into an accelerating growth trajectory.

This white paper is intended for CEOs, CFOs, and strategic decision-makers navigating AI transformation. We hope it serves as a practical decision-support tool — not a forecast of the future, but a means of clearly understanding where your organization stands today.

01 Framework Definition and Indicator Classification of HRI

Hedge Resilience Index (HRI) is not simply another AI scoring tool, it is a quantitative measure of an enterprise's capacity to sustain its AI transformation.

In the AI era, every enterprise faces the same fundamental challenge: legacy revenues are declining whilst AI-driven revenues are in early ascent. The critical question is whether available cash and time are sufficient to complete the transformation before legacy operations bottom out. HRI translates this challenge into a set of quantifiable, trackable, and cross-comparably benchmarkable indicators.

The central premise of HRI is that AI transformation is not primarily a technology challenge — it is a "hedging" challenge. Enterprises must use the growth velocity of AI-driven revenues to hedge against the decline of legacy revenues, and use the anticipated future value of AI investments to offset current capital consumption. Whether this hedging succeeds depends on the combined performance across seven Indicators.

The seven indicators span five dimensions — business structure, growth velocity, future value, capital efficiency, and strategic time — collectively providing a comprehensive view of an enterprise's AI transformation resilience.

1.1 Seven Indicators • Five Dimensions

Table 1-1 The Seven Hedge Resilience Index (HRI) Indicators and Their Core Questions

Dimension	Indicator Name	Abbreviation	Key Question Addressed
Business Structure	Raw Hedge Ratio	HRR	Can current AI revenues offset the legacy revenue gap?
Growth Velocity	Hedge Acceleration Coefficient	HAC	Is the AI engine growing fast enough?
	Augmented Acceleration Coefficient	HAC-Aug	What is the amplification effect of AI across total operations?
Future Value	Hedge Operational Option Value	HRI-OP	What is the long-term value of transforming now?
	Contract Hedge Index	HRI-C	How much future revenue is locked in by contracts?
Capital Efficiency	Capital Efficiency Index	HRI-CapEx	What is the AI return on each unit of CapEx?
Strategic Time	Strategic Window Period	T_Window	How much strategic window remains?

1.2 HRI Indicator Composition

The HRI framework comprises 7 core scoring indicators:

7 core scoring indicators (constituting the HRI composite score of 0–100)

HRR, HAC, HAC-Aug, HRI-OP, HRI-C, HRI-CapEx, T_Window

1.3 Data Sources and Scope

HRI scores are calculated using publicly available data sources. The table below outlines the scope of data collection.

Table 1-2 Scope of Data Collection for the Hedge Resilience Index (HRI)

Category	Description
Data Sources	Statutory disclosure filings by listed companies, official corporate publications, financial data from specialist providers, research reports by major investment banks, and credible published news sources
Coverage	The HRI framework is applicable to publicly listed enterprises across sectors.

02 The Value and Significance of HRI

In the AI era, each stakeholder faces distinct blind spots — the CEO is concerned with the urgency of the strategic window; the CFO with cash runway; the CTO with where the critical capability gaps lie. HRI provides a common analytical framework across all these perspectives.

2.1 For Investors: From Narrative to Analytical Positioning

AI narratives have become pervasive. Nearly every listed company presents an AI transformation story, yet distinguishing genuine progress from superficial positioning is difficult — definitions of AI revenue vary, corporate disclosures are self-reported, and reporting standards remain inconsistent. Investors face three core challenges:

Table 2-1 Three Core Challenges Facing Investors and the Value Delivered by the Hedge Resilience Index (HRI)

Investor Challenge	Value Provided by HRI
Authenticity	Is this genuine AI transformation or AI-washing? HRI applies a rigorous AI revenue classification standard that excludes non-genuine AI revenues, enabling every company to be assessed against a consistent benchmark.

	The AI Wash Index (AI_Wash_Index) triggers an automatic alert when the designated threshold is exceeded.
Timing	Is this the right time to invest? The composite score maps to one of five positions on the U-curve, enabling rapid identification of where a company stands in its AI transformation — value trap or pre-reversal stage, peak performance or potential winner's curse.
Comparability	Company A vs Company B: which demonstrates greater AI transformation resilience? The seven indicators allow enterprises across different sectors, scales, and stages of development to be assessed against a common standard.

2.2 For Enterprise Leaders: From Intuition to Analytical Grounding

Different C-suite roles have fundamentally different priorities. HRI provides tailored diagnostic dimensions for six core executive roles — not a single standardized answer, but a means of identifying the specific constraints most relevant to each leader.

Table 2-2 Core Concerns of Different Enterprise Operators and the Value Delivered by the Hedge Resilience Index (HRI)

Role	Primary Concerns	Value Provided by HRI
CEO	Overall strategic direction; critical transformation junctures	U-curve positioning combined with crossover point projection, providing clarity on whether the enterprise is within its strategic window or approaching a critical inflection point.
COO	Operational efficiency; depth of AI integration in internal processes	HAC-Aug quantifies the amplification effect of AI across total operations. A low reading indicates that AI remains in isolated deployment rather than generating systemic operational uplift.
CMO	Market positioning; integrity of AI revenue claims and external communications	HRI's AI revenue classification standard distinguishes between AI-native, AI-transformation, and AI-augmented revenue, enabling brand positioning to align with verifiable operational capabilities and reducing the risk of AI-washing in external communications.
CFO	Capital allocation; cash runway; capital constraints at the crossover point	The crossover-capital constraint coupling model quantifies the balance between capital and time — when the projected crossover point under the pessimistic scenario exceeds the capital depletion horizon, an emergency response protocol is triggered.
CSO	Industry positioning; paradigm classification; competitive benchmarking	Post-calculation paradigm clustering combined with sector HRI rankings provides clear strategic positioning — whether the trajectory most resembles Meta or Oracle, the appropriate path differs substantially.
CTO	Return on AI technology investment; technology	The seven-indicator radar chart precisely identifies capability gaps — whether AI revenue growth is

	capability gaps	insufficient (low HRR/HAC) or capital investment has not yet translated into measurable outcomes (low HRI-CapEx).
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03 The U-Curve: Where Does Your Enterprise Stand?

The U-Curve is a visual representation of the Hedge Resilience Index (HRI). The horizontal axis represents the HRI composite score (0–100); the vertical axis represents the conceptual position of enterprise value.

The left side (including Transformation Laggard Zone and Left-Side Decline) describes enterprises whose legacy operations remain substantial in scale but whose transformation is insufficient; the right side (including Peak Performance and Right-Side Acceleration) describes those where AI-related business has reached meaningful scale; the middle — the "U-Curve Trough" — is the pressure phase of the old-to-new business transition. Enterprises at the trough may continue under pressure, or may resume growth after key adjustments. The direction is determined not only by the industry environment, but also by how management allocates capital, reshapes the business portfolio, and drives organizational execution.

It should be noted that the U-Curve is a conceptual positioning map based on the HRI framework — not a market-capitalization or time-based prediction model.

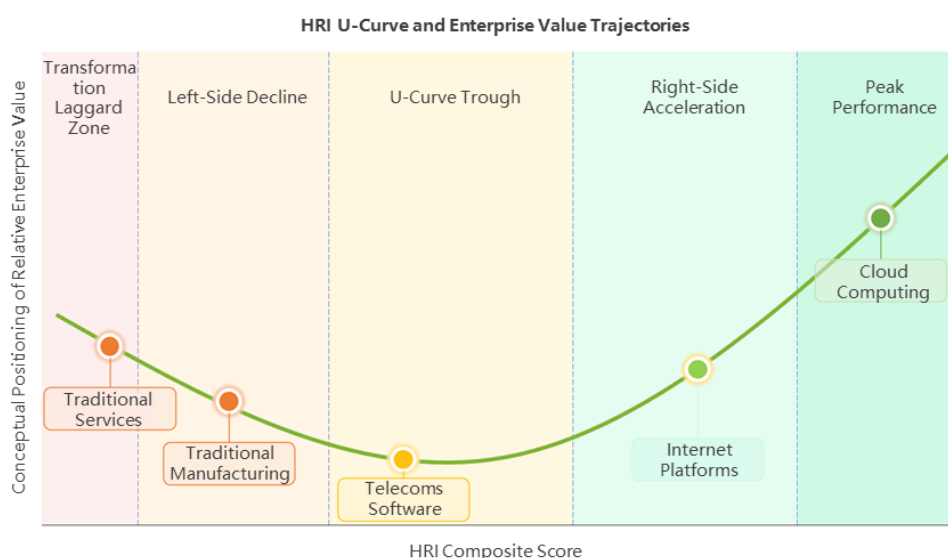


Figure 3-1 HRI U-Curve: Five Strategic Positions and Enterprise Value Trajectories (Illustrative)

Table 3-1 The HRI U-Curve: Five Positions and Their Strategic Implications

U-Curve Position	Strategic Implication	Indicative Sector / Type
Peak Performance	AI capabilities have reached a leading level of maturity; significant competitive advantages established	Cloud computing
Right-Side Acceleration	Crossover point completed; AI-driven growth accelerating	Internet platforms
U-Curve Trough	Critical transformation phase; acceleration required	Telecoms software
Left-Side Decline	AI foundation exists but has not yet generated effective hedging	Traditional manufacturing
Transformation Laggard Zone	Transformation severely insufficient; risk of strategic falling behind	Traditional services sector

04 Five Paradigms: Which Paradigm Applies to Your Enterprise?

4.1 Why Are Paradigms Necessary?

GE, IBM, NVIDIA, Baidu, Meta, and Oracle are each navigating fundamentally different transformation trajectories. Meta is not hedging against decline — it is amplifying growth. Oracle is not hedging through current revenues — it is locking in future revenues through contractual commitments. NVIDIA did not rely on operational efficiency; it relied on capital efficiency to sustain itself through a period of large-scale investment ahead of revenue materialization. Assessing these companies' transformation performance purely through financial ratios, without recognizing their respective transformation logic, would inevitably produce misleading conclusions. This is precisely the design rationale for HRI: the framework does not measure whether transformation is good or bad in absolute terms, but rather where you currently stand in terms of transformation trajectory and paradigm positioning.

Two enterprises with an identical HRI score of 52 may be pursuing diametrically opposite strategies — one a growth amplifier, the other hedging against decline. Enterprises with equal AI investment may follow entirely different paths: one locking in revenue through contracts, another building compute infrastructure. Without paradigm classification, it is impossible to explain why equal scores can correspond to such different outcomes.

4.2 What Is an AI Transformation Paradigm?

An AI Transformation Paradigm is an analytical classification of the core constraint types and transformation logic characteristics that enterprises face in the AI era. It is a high-level characterization of an enterprise's AI revenue sources, cash flow structure, capital dependency, and risk profile — used to precisely describe the underlying transformation model, identify actual constraints and bottlenecks, and provide matched strategic guidance.

A paradigm is not an externally assigned label — it is a clustering of structural characteristics that emerge naturally from an enterprise's AI transformation journey. It answers one central question: what underlying logic does your enterprise rely upon to achieve AI hedging and organizational continuity?

HRI paradigms have three key characteristics: (1) Paradigm is a result, not an input — the seven indicators are calculated first, then paradigm type is derived through radar chart feature clustering; it does not pre-determine weightings or affect scores; (2) An enterprise has only one primary paradigm at any given time, avoiding analytical ambiguity; (3) Paradigms are not used in calculation — only for diagnostic profiling, interpretation, and strategic guidance matching. Paradigms reflect transformation patterns rather than sector, scale, or national differences.

Paradigm I Crossover Hedge

"Approaching a critical inflection point — either at the golden window or near the edge of strategic failure."

Legacy operations are in irreversible decline whilst AI-native businesses are in early stages. The enterprise is approaching a critical crossover threshold. If AI revenue growth is sufficiently rapid, the crossover can be completed before legacy revenues collapse; if not, the enterprise may slide towards the transformation laggard zone.

For these enterprises, there is no "maintain the status quo" option — either accelerate the crossover or face structural decline.

Table 4-1 Paradigm I · Crossover Hedge — Core Indicator Characteristics and Strategic Priorities

Key Indicator Profile	AI revenues are small in scale but growing rapidly; time window is constrained
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Illustrative Examples	Traditional IT outsourcing and software service providers whose traditional core business faces structural substitution and are transforming with new AI businesses as a hedge.
Strategic Priority	Shortening the time to crossover is the single critical priority.

Paradigm II Amplification-Led Growth

"Dual-engine growth, with no existential threat on the horizon but competitive pressure in the near term — insufficient AI amplification risks falling behind."

Core business continues to grow healthily. AI is not deployed to hedge against decline, but to amplify growth. AI is deeply embedded in core products, enhancing user experience, conversion rates, and average revenue per user.

There is no current existential risk, but competitive pressure is present — if competitors achieve greater AI amplification, market share may be eroded. The challenge is near-term competitive, not long-term structural.

Table 4-2 Paradigm II · Amplification-Led Growth — Core Indicator Characteristics and Strategic Priorities

Key Indicator Profile	AI revenues and returns have reached meaningful scale, with a more comfortable strategic time window
Illustrative Examples	Tech companies with healthy existing core businesses that can achieve value-added growth through AI feature upgrades (e.g., companies seeing rising cloud service revenue alongside rapidly scaling AI service revenue).
Strategic Priority	Continuously improve HAC-Aug; expand the amplification effect of AI across operations.

Paradigm III Contract-Secured

"Temporary runway secured by existing contracts — but the visible backlog is finite."

Revenue structure is dominated by long-term contracts (e.g. B2B cloud computing, industrial software, AI infrastructure services). These contracts lock in 1–3 years of future revenues, providing a meaningful buffer period.

Contracts are a double-edged instrument — the existing backlog is being consumed, and the pace of new contract signings determines the future trajectory. For these enterprises, the risk lies not in the present but in the foreseeable future.

Table 4-3 Paradigm III · Contract-Secured — Core Indicator Characteristics and Strategic Priorities

Key Indicator Profile	Stable long-term contracts provide visibility over future revenues
Illustrative Examples	Telecom operators, large enterprise software companies (with multi-year licensing agreements already signed), and infrastructure concession enterprises that have multi-year service contracts.
Strategic Priority	Accelerate new contract signings; expand remaining performance obligation (RPO) backlog.

Paradigm IV Capital-Intensive

"A current focus of AI investment attention in capital markets, though elevated valuation risk may be present."

CapEx as a proportion of revenues is very high (typically >50%), with large-scale capital expenditure directed at AI infrastructure (data centres, GPU clusters, chip R&D). These enterprises occupy the physical foundation layer of the AI value chain.

Market valuations reflect high expectations, but the risk is clear: if capital efficiency (HRI-CapEx) does not improve, or if revenue growth fails to keep pace with CapEx expansion, valuation multiples may compress significantly.

Table 4-4 Paradigm IV · Capital-Intensive — Core Indicator Characteristics and Strategic Priorities

Key Indicator Profile	Revenue growth is driven by substantial investment, to some extent reflecting market confidence in the business model
Illustrative Examples	AI chip design companies, large-scale intelligent computing center operators, and tech giants that have built their own GPU clusters for large model training.
Strategic Priority	Improve HRI-CapEx; demonstrate the return on capital investment.

Paradigm V Efficiency Internalization

"Using AI to reduce costs, but not yet using AI to generate revenues."

AI capabilities have not yet been converted into direct external revenue. Instead, AI is deployed as an internal tool for cost reduction and efficiency improvement. AI investment is directed primarily at enhancing operational efficiency, reducing unit costs, and improving decision quality.

Unlike the other four paradigms, the AI value of Paradigm V enterprises sits on the cost side of the income statement rather than the revenue side. The key question is whether the enterprise can convert its cost advantage into a market or revenue

advantage within the available window.

Table 4-5 Paradigm V · Efficiency Internalization — Core Indicator Characteristics and Strategic Priorities

Key Indicator Profile	Direct AI revenues are low; efficiency-related savings are substantial; legacy business is in steady-state transformation
Illustrative Examples	Enterprises in traditional manufacturing (AI quality inspection, predictive maintenance), retail (AI supply chain optimization, smart inventory management), and financial services (AI automated financial reporting, intelligent risk control) where internal efficiency improvement is the primary path to realizing AI value.
Strategic Priority	Convert efficiency advantages into productisable AI capabilities within the strategic window. Typical migration path: Paradigm V → Paradigm II.

05 The Crossover Point: The Capital Constraint That Determines Outcomes

5.1 What Is the Crossover Point?

Every enterprise undergoing AI transformation is simultaneously managing two revenue curves — the decline curve of legacy business revenue and the growth curve of AI-driven revenue. The Crossover Point is the moment at which AI revenue growth first compensates for, and then surpasses, the decline in legacy revenues. This is the most central single concept in the HRI framework, and the most critical time horizon for a CEO to understand.

Prior to the crossover point, the enterprise is in a cash consumption phase — legacy revenues are contracting, AI revenues are insufficient to fill the gap, and the enterprise depends on capital reserves to sustain the transformation. After the crossover point, AI revenues begin to drive growth and the enterprise enters a self-sustaining positive cycle. The timing of the crossover point directly determines whether the enterprise enters the right-side acceleration trajectory or slides into the transformation laggard zone.

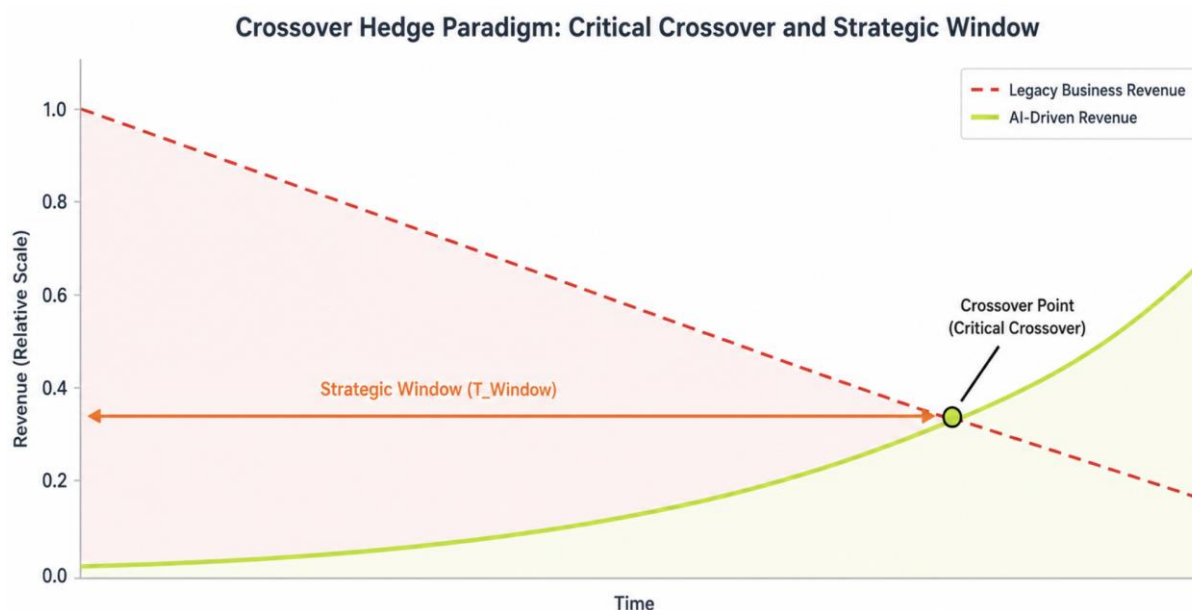


Figure 5-1 Schematic Diagram of the AI Revenue Crossover Point

5.2 The Crossover Point-Capital Constraint Coupling

The real challenge of the crossover point is not "will it come" — it is "can capital reserves sustain the enterprise until it does." HRI couples the projected crossover timeline with the capital depletion timeline, identifying a CEO-level emergency trigger condition. The core inequality for this coupling is:

$T_{\text{Crossover}}$ (projected crossover point under pessimistic scenario) $>$ $T_{\text{CashRunout}}$ (capital depletion time at current CapEx pace) When this inequality holds — meaning capital will be exhausted before the crossover point is reached — the enterprise is in a zone of significant operational risk. This is the trigger condition within the HRI framework that requires immediate CEO-level action rather than waiting for the next quarterly update.

IBM's case offers a particularly instructive illustration. In 2015, IBM was positioned at the U-curve trough with legacy revenues in sustained decline. In 2019, the USD 34 billion acquisition of Red Hat led by Arvind Krishna became a strategic turning point — HRI showed a notable improvement in the year following the acquisition, and the projected strategic window extended from 22 months to 30 months. By 2026, IBM's share price had reached new highs. The journey from the U-curve trough to a more stable position took eleven years.

Table 5-1 Crossover Point-Capital Constraint Coupling: Assessment Scenarios and Recommended Actions

Coupling Assessment Scenarios	Status	Recommended Action
Baseline crossover time < capital depletion time $\times 0.6$	Safe Zone	Proceed with standard transformation pace
Baseline crossover time $\approx$ capital depletion time $\times 0.8$	Watch Zone	Track capital efficiency improvement trends on a quarterly basis
Pessimistic crossover time > capital depletion time	Critical Zone	Immediate CEO-level response required; initiate strategic restructuring or capital raising without delay

06 Your Next Steps

The legacy vessel is taking on water; the AI-era ship is preparing to launch. The question is whether your available capital, time, and organizational capability are sufficient to secure a place on board.

This is not rhetoric — it is the core mission of HRI: to help you answer this question with precision. Wherever your enterprise stands on the U-curve, HRI provides clear positioning and actionable guidance.

| For the CEO

If your enterprise is in the U-curve trough, this is the critical strategic decision period. Accelerate investment or pursue strategic retrenchment? HRI provides data-driven support rather than intuition-based judgement. Both IBM and GE demonstrate that the U-curve trough is not a destination — it is the starting point for strategic choice.

| For the COO

HAC-Aug quantifies the amplification effect of AI across total operations. If your enterprise's HAC-Aug is low, it indicates that AI remains in isolated deployment, with no systemic amplification effect. Consider: which operational processes most warrant AI-driven uplift, and which investments may be misallocated towards short-term cost reduction rather than transformational impact?

| For the CMO

Be mindful of the three revenue categories defined by the HRI AI revenue classification standard. If externally communicated AI revenues are substantially

derived from AI-washing or internal efficiency savings alone, the AI_Wash_Index will trigger a warning — and the market will eventually apply a longer evaluation window to assess genuine progress. Build marketing narratives on verifiable AI-native and AI-transformation-driven revenues.

| For the CFO

Calculate your strategic window. The balance between cash reserves and transformation velocity is the most critical financial decision you face. The crossover-capital constraint coupling model quantifies this balance — when the projected crossover point under the pessimistic scenario exceeds capital depletion time, a CEO-level emergency response is triggered. This is the only condition within the HRI framework that cannot wait for the next quarterly update.

| For the CSO

Paradigm classification determines strategic direction. First confirm which of the five paradigms applies to your enterprise, then determine whether the priority is "accelerating the crossover" (Paradigm I Crossover Hedge), "amplifying advantages" (Paradigm II Amplification-Led Growth), "expanding contract backlog" (Paradigm III Contract-Secured), "demonstrating ROI" (Paradigm IV Capital-Intensive), or "converting cost savings into revenue generation" (Paradigm V Efficiency Internalization). The same strategic action produces very different outcomes depending on paradigm.

| For the CTO

The seven-indicator radar chart will precisely identify where your capability gaps lie. Is AI revenue growth insufficient (low HRR / HAC)? Or has AI investment not yet translated into measurable outcomes (low HRI-CapEx)? The indicators do not mislead. Technology investment priorities should be driven by where the radar chart shows weakness — not by where market attention happens to be focused.

| For Investors

Move beyond AI narratives and begin examining HRI positioning. Whether a company is at peak performance, right-side acceleration, or in the U-curve trough determines whether the appropriate analytical perspective should be "stable equilibrium" or "growth premium." An AI Wash Index exceeding three times the benchmark may indicate material downside risk — this should always be considered

alongside sector benchmarks and underlying fundamentals.

The first wave of AI-era winners will not be those with the most powerful technology — they will be those who most clearly understand where they stand and how fast they need to move.

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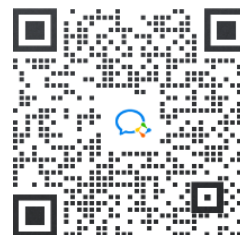
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